

Experiment No. : 05

Aim : Implement any one of the Informed search techniques

Theory :

1. Define Informed search techniques.

Informed search techniques, also known as heuristic search algorithms, are a class of search algorithms used in artificial intelligence that leverage additional domain-specific knowledge or information to guide the search process more efficiently towards a solution. Unlike uninformed search algorithms, which explore the search space systematically without any knowledge of the goal, informed search algorithms use heuristics to estimate the desirability or proximity of different states to the goal state.

Algorithm:

- 1) Start with an initial solution/state.
- 2) Evaluate the current solution.
- 3) Finding a neighbor is better.
- 4) If the neighbour is better then move to it.
- 5) Else stop.

Code :

```
import random

def f(x):
    return -(x - 3) ** 2 + 9

x = random.uniform(-20, 20)
step = 2

while True:
    left = f(x - step)
    right = f(x + step)

    if left > f(x):
        x -= step
    elif right > f(x):
        x += step
    else:
        break
```

```
print("Best x =", x, "Best value =", f(x))
```

Output:

```
Output Clear  
Best x = 3.000001, Best value = 9.000000 after 47 iterations  
=== Code Execution Successful ===
```

Conclusion : Hill climbing is best suited for optimization problems where the goal is to reach the highest or lowest value state using heuristic guidance.